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Testing Document and Specification for Budget Bridge Web App

Test Plan

Introduction

This document outlines the test plan for the Budget Bridge web application. As outlined in the project Requirements Document, this system will provide a personal budgeting application for users to use to track their income and expenses. The testing activities discussed in this document will verify that the web application meets the needs of the Commerce Bank team, our professor for University of Missouri—Kansas City's (UMKC) Software Engineering Capstone (CS-451R) course and future users by verifying that the requirements for this web application, as outlined in the Requirements Document, are met.

The testing routine will test Budget Bridge's functions that incorporate both the front-end and back-end parts of the web application. The results of this testing procedure will give the developers of Budget Bridge assurance that the software works as intended to.

Test Objectives

1. Ensure that data is able to be sent frontend between the backend with no issue throughout the entire time of a user session
2. Confirm that the different pages of the web application render the correct content from the backend and MySQL database.
3. Ensure that the login/registration functionality properly affects what data the frontend renders from the data it receives from the backend.

Testing Strategy

We will employ automated testing with the test tools below in order to verify the frontend and backend functionalities work as intended. We're looking to achieve 10% code coverage.

Test Tools and Resources

- JUnit 5 (<https://junit.org/>)

Items Tested (Scope)

Items that will be tested during the testing phase as laid out by the Requirement Document will be but are not limited to:

- Ability for an individual to login and register for an account on Budget Bridge
Test Case: 4.1.1
- Ability for a user to use the functionality of the home page/dashboard after the user has logged
Test Case: 4.1.2
- Ability for a user to input expenses in the Spending page
Test Case: 4.1.3
- Ability for a user to input income in the Income page
Test Case: 4.1.4
- Ability to a user to access the functionality in the User Profile / Settings page
Test Case: 4.1.5

Test Deliverables

The Test Plan, Test Specification, and the Test Report are documents that are linked. The chart below shows how they relate to each other.

Deliverables	Description
Test Plan	This document outlines the overarching objectives we have for testing and how we will go about testing our web application.
Test Specifications	This document outlines what is included in each test case.
Test reports	This document outlines the results of our tests that we did on our web application.